



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

Mathematical Foundations of Data Science (Probability)

Lecture 06

Karthik P. N.

Assistant Professor, Department of AI

Email: pnkarthik@ai.iith.ac.in

26 September 2026

IID Random Variables

Independent and Identically Distributed (IID) Random Variables

We say that $X_1, \dots, X_n$ are **independent and identically distributed (IID)** if:

- $X_1, \dots, X_n$ are **mutually independent**.
- $X_1, \dots, X_n$ **have the same CDF**, i.e., $F_{X_1} = \dots = F_{X_n}$.

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Remarks:

- If $X_1, \dots, X_n$ are jointly discrete, then the phrase “**have the same CDF**” can be replaced with “**have the same PMF**”, i.e.,

$$p_{X_1} = \dots = p_{X_n}.$$

- If $X_1, \dots, X_n$ are jointly continuous, then the phrase “**have the same CDF**” can be replaced with “**have the same PDF**”, i.e.,

$$f_{X_1} = \dots = f_{X_n}.$$

Maximum of Random Variables

Fix $(\Omega, \mathcal{F}, \mathbb{P})$.

Let $X_1, \dots, X_n$ be random variables defined with respect to $\mathcal{F}$, with joint CDF $F_{X_1, \dots, X_n}$.

- Show that $Y_n = \max\{X_1, \dots, X_n\}$ is a random variable with respect to $\mathcal{F}$.
- Derive the CDF of Y_n .
- Simplify the CDF of Y_n when $X_1, \dots, X_n$ are IID.

Minimum of Random Variables

Fix $(\Omega, \mathcal{F}, \mathbb{P})$.

Let $X_1, \dots, X_n$ be random variables defined with respect to $\mathcal{F}$, with joint CDF $F_{X_1, \dots, X_n}$.

- Show that $Z_n = \min\{X_1, \dots, X_n\}$ is a random variable with respect to $\mathcal{F}$.
- Derive the CDF of Z_n .
- Simplify the CDF of Z_n when $X_1, \dots, X_n$ are IID.

Minimum of i.i.d. Exponential Random Variables

Fix $(\Omega, \mathcal{F}, \mathbb{P})$. Let $\lambda_1, \dots, \lambda_n > 0$ be fixed constants.

Let $X_1, \dots, X_n$ be independent, with $X_i \sim \text{Exponential}(\lambda_i)$ for each $i \in \{1, \dots, n\}$.

Find the distribution of $Z = \min\{X_1, \dots, X_n\}$.

Sums of Random Variables

Fix $(\Omega, \mathcal{F}, \mathbb{P})$.

Let X and Y be random variables with respect to $\mathcal{F}$.

- Show that $X + Y$ is a random variable with respect to $\mathcal{F}$.
- In the cases when X and Y are jointly discrete/continuous, derive the PMF/PDF of $X + Y$.
- Simplify the PMF/PDF when X and Y are independent.

General Transformations and the Jacobian Formula

General Transformations

Fix $(\Omega, \mathcal{F}, \mathbb{P})$.

Let $X : \Omega \rightarrow \mathbb{R}$ be a **continuous random variable** with PDF f_X .

Suppose that we are given $g : \mathbb{R} \rightarrow \mathbb{R}$ that:

1. Is **(strictly) monotone and differentiable** throughout its domain.
2. Has **non-zero derivative everywhere** in its domain, i.e., $g'(x) \neq 0$ for all $x \in \mathbb{R}$.

What is the PDF of $Y = g(X)$?

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Note that because g is strictly monotone, admits an inverse, say $g^{-1} : \mathbb{R} \rightarrow \mathbb{R}$

Case 1: g Strictly Increasing

$$\begin{aligned}F_Y(y) &= \mathbb{P}(\{Y \leq y\}) \\&= \mathbb{P}(\{g(X) \leq y\})\end{aligned}$$

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Thus, we have

$$f_Y(y) = \begin{cases} \frac{f_X(g^{-1}(y))}{g'(g^{-1}(y))}, & y \in \text{Range}(g), \\ 0, & y \notin \text{Range}(g). \end{cases}$$